Positive solutions of the Felder–Hoppe recurrences for quantum minimal surfaces

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Abstract. For every integer $d \geq 2$ and every $a > 0$, we prove the conjecture of Felder and Hoppe that the leading Gram matrices of their indefinite Hermitian form with density $\exp\bigl(-a|z|^2 + \frac{\mathrm{i}}{d}(z^d + \bar{z}^d)\bigr)$ are nondegenerate, and we show that all signed norms of the corresponding orthogonal polynomials are positive. Consequently, the discrete Painlevé-type recurrence satisfied by the norm ratios, which arises from the quantum minimal surface $w_2 = w_1^{d-1}$, has a positive solution with the moment initial data of Felder and Hoppe. The proof writes the planar integral that generates each symmetry block as the Laplace transform of a product of $d-1$ independent Gamma variables. The Gram blocks become Hankel matrices of these Laplace transforms, and total positivity of the logarithmic Gamma kernels, together with Andréief's identity, fixes the sign of every leading minor. For $d = 4$ we identify the initial data with Bessel functions of orders $\pm\frac{1}{4}$. Finally, for every $d \geq 3$ we show that the positive solution is unique and determine its growth, by comparing an arbitrary positive solution with an explicit approximate solution through the convex variational structure of the recurrence.

Keywords. discrete Painlevé-type recurrence; indefinite orthogonal polynomials; total positivity; Gamma distribution; quantum minimal surface.

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1. Introduction

Discrete Painlevé-type recurrences occur both in matrix regularizations of minimal surfaces and as recurrences for the coefficients of orthogonal polynomials. For the quantum minimal surfaces associated with the holomorphic curves $w_2 = w_1^{d-1}$, recurrences of this type were obtained in [CT99, AHK19, Hop25] (see [FH25, Section 3.1]). The quartic recurrence is written explicitly in [Hop25, equation (38)] and [CDHM25, Section 5]; those works also treat the positive special-function solution in the cubic case $d = 3$.

Felder and Hoppe [FH25] constructed an indefinite Hermitian form for each d (recalled in Section 2) and showed that, whenever its leading Gram matrices are nondegenerate, the

ratios V_n of its orthogonal-polynomial norms obey

$$V_n + \frac{1}{a^2}\sum_{j=0}^{d-2}\prod_{k=0}^{d-2} V_{n+j-k} = \frac{n+1}{a}, \qquad n \geq 0, \tag{1.1}$$

where $V_n = 0$ for $n < 0$, with the initial values

$$V_j = \frac{\int_{\mathbb{C}} |z|^{2j+2}\rho_a \,\mathrm{d}A}{\int_{\mathbb{C}} |z|^{2j}\rho_a \,\mathrm{d}A}, \qquad 0 \leq j \leq d-3, \tag{1.2}$$

where ρ_a is the density (2.1) [FH25, equation (3.3) and the display after it]. They proved the required Gram nondegeneracy for $d = 1, 2, 3$ and conjectured it for every d [FH25, Conjecture 3.6 and Theorem 3.7]. Their conjecture is stated for $a \geq 0$. We treat $a > 0$, where all planar moments converge absolutely; at $a = 0$ they do not, and the endpoint would require an oscillatory or regularized interpretation, which we do not discuss.

Theorem 1.1. *Let $d \geq 2$ and $a > 0$.*

(i) *Every leading Gram matrix of the Felder–Hoppe Hermitian form* (Section 2) *is nondegenerate, and all its signed norms $\mathfrak{h}_n$ are positive. Thus [FH25, Conjecture 3.6] holds for every d and every $a > 0$.*

(ii) *Let $d \geq 3$. The ratios $V_n = \mathfrak{h}_{n+1}/\mathfrak{h}_n$ form the unique real sequence that satisfies* (1.1) *for every $n \geq 0$ and has the initial values* (1.2)*. In particular, this solution is positive.*

The moments in (1.2) are explicit expectations of Gamma variables (Proposition 4.1). For $d = 4$, after setting $v_n = V_n/a$ and $\epsilon = a^{-2}$, the recurrence (1.1) becomes

$$v_n\,(1 + v_{n-1}v_{n-2} + v_{n+1}v_{n-1} + v_{n+2}v_{n+1}) = (n+1)\epsilon, \qquad n \geq 0, \tag{1.3}$$

where $v_n = 0$ for $n < 0$. The two initial values are

$$v_0 = -\frac{1}{a}\frac{h'(a)}{h(a)}, \qquad v_1 = -\frac{1}{a}\frac{h''(a)}{h'(a)}, \tag{1.4}$$

with

$$\begin{aligned} h(a) = \frac{\pi^2 a}{4}\Bigg[& J_{-1/4}\left(\frac{a^2}{4}\right)^2 - \sqrt{2}\, J_{-1/4}\left(\frac{a^2}{4}\right) J_{1/4}\left(\frac{a^2}{4}\right) \\ & + J_{1/4}\left(\frac{a^2}{4}\right)^2 \Bigg]. \end{aligned} \tag{1.5}$$

This is the function $h(a) = \int_{\mathbb{C}} \rho_a \,\mathrm{d}A$ of [FH25, equation (3.4)] for $d = 4$, in the Bessel form given in [FH25, Section 4]; Lemma 7.2 below rederives it.

Theorem 1.2. *Let $a > 0$. The initial values* (1.4)–(1.5) *are well defined and extend uniquely to a real sequence satisfying* (1.3) *for every $n \geq 0$. This sequence is $v_n = \mathfrak{h}_{n+1}/(a\mathfrak{h}_n)$ for $d = 4$; in particular, $v_n(a) > 0$ for every $n \geq 0$.*

Uniqueness in Theorems 1.1(ii) and 1.2 refers to the prescribed initial data. For $d \geq 3$, the substitution $V_n = a^{2/(d-2)} v_n$ and $\epsilon = a^{-d/(d-2)}$ turns (1.1) into

$$v_n + \sum_{m=n-d+2}^{n} v_m v_{m+1} \cdots v_{m+d-2} = (n+1)\epsilon, \qquad n \geq 0, \tag{1.6}$$

where $v_n = 0$ for $n < 0$; for $d = 4$ this is (1.3). The next theorem shows that the positive solution itself is unique, so that the moment values (1.2), and for $d = 4$ the Bessel values (1.4), are the only initial values leading to a positive solution.

Theorem 1.3. *Let $d \geq 3$ and $\epsilon > 0$. Every sequence of positive numbers $(v_n)_{n \geq 0}$ satisfying (1.6) for every $n \geq 0$ satisfies*

$$\lim_{n \to \infty} v_n \left(\frac{d-1}{\epsilon(n+1)} \right)^{1/(d-1)} = 1,$$

and there is at most one such sequence. In particular, for $a > 0$ the sequence (V_n) of Theorem 1.1(ii) is the unique positive solution of (1.1), and for $d = 4$ the sequence of Theorem 1.2 is the unique positive solution of (1.3).

For the cubic analogue, the existence of a positive solution was proved in [AHK19, Section 7] and its uniqueness in [CDHM25, Theorem 2.1, proved in Section 4.2]; for $d = 3$, Theorem 1.3 gives a short proof of that uniqueness. For the quartic recurrence, [CDHM25, Section 5] reports preliminary evidence for a positive solution with analogous properties. Our proof of Theorem 1.3 (Section 8) uses only the recurrence and is independent of the rest of the paper. In logarithmic variables the recurrence is the stationarity condition of a convex functional. Comparing an arbitrary positive solution with an explicit approximate solution, whose residual is $O(n^{-1})$, gives an a priori bound in which the convexity is controlled by the known approximate solution rather than by the unknown one, and a cutoff argument then compares two positive solutions.

The proof of Theorem 1.1(i) treats all d and all indices at once. Independent source parameters u and v for z^d and $\bar{z}^d$ are used on disks, where the two-source integral is entire and, by rotation invariance, depends only on uv. In exponential coordinates $u = \mathrm{e}^{t_1}$, $v = \mathrm{e}^{t_2}$ this turns the Gram block of each residue class modulo d, after a unitriangular Stirling transformation, into the Hankel matrix of derivatives of a single kernel $F_r(t)$, the planar integral with weight $|z|^{2r}$ and coupling $\mathrm{e}^{t/2}$ (Lemma 3.1). A Mellin transform and Gauss's multiplication formula show that F_r is the Laplace transform, in e^t, of a product of $d-1$ independent Gamma variables (Proposition 4.1). The signs of its Hankel determinants then follow from total positivity: the densities of logarithms of Gamma variables are totally positive convolution kernels, and Andréief's identity writes each Hankel determinant as an integral of a product of two determinants of fixed sign (Lemma 5.1). This determinant step is a continuous Cauchy–Binet argument of the kind used for biorthogonal ensembles [Bor99]. To our knowledge, the new ingredients are the two-source representation, which separates the insertions of z^d and $\bar{z}^d$ that a common coupling mixes, and the Gamma-product form of the kernels.

Section 2 recalls the Hermitian form. Section 3 reduces the Gram blocks to Hankel determinants, Section 4 proves the Gamma representation, and Section 5 proves Theorem 1.1(i). Section 6 derives the recurrence and proves Theorem 1.1(ii). Section 7 treats $d = 4$ and proves Theorem 1.2, and Section 8 proves Theorem 1.3. For $d = 4$, Theorems 1.1, 1.2 and 1.3 have been formally verified in Lean 4.33.0 with Mathlib, as the declarations `Painleve.main_theorem` and `Painleve.unique_theorem`. That formalization follows earlier, quartic-specific forms of the arguments (see the code availability statement); the general-d arguments of Sections 3–6 and 8 have not been formalized.

2. The Gram form

Fix $d \geq 2$ and $a > 0$, and put

$$\rho_a(z) = \exp\left(-a|z|^2 + \frac{\mathrm{i}}{d}(z^d + \bar{z}^d)\right), \tag{2.1}$$

the density of [FH25, Section 3.1]. The Gaussian factor makes all polynomial moments absolutely convergent. Let $\eta = \mathrm{e}^{\pi \mathrm{i}/d}$. Since $\eta^d = -1$,

$$\rho_a(\zeta z) = \rho_a(z) \quad (\zeta^d = 1), \qquad \rho_a(\eta z) = \overline{\rho_a(z)}.$$

Decompose the polynomial ring into weight spaces

$$\mathbb{C}[z] = \bigoplus_{r=0}^{d-1} \mathcal{P}_r, \qquad \mathcal{P}_r = \{f \in \mathbb{C}[z] : f(\zeta z) = \zeta^r f(z) \text{ for } \zeta^d = 1\} = \operatorname{span}\{z^{r+dq} : q \geq 0\}.$$

For $f \in \mathcal{P}_r$, set $(\sigma f)(z) = (-\eta)^{-r} f(\eta z)$, extend σ linearly to $\mathbb{C}[z]$, and define

$$(f, g)_a = \int_{\mathbb{C}} f(z)\overline{\sigma g(z)}\rho_a(z)\,\mathrm{d}A(z), \qquad \mathrm{d}A = \mathrm{d}(\operatorname{Re} z)\,\mathrm{d}(\operatorname{Im} z), \tag{2.2}$$

which is linear in f and conjugate-linear in g. This is the Hermitian form of [FH25, Proposition 3.1], with the right inverse $\chi(r) = r$ $(0 \leq r \leq d-1)$ chosen there. If $m = r + dq$ with $0 \leq r \leq d-1$, then

$$\sigma z^m = (-\eta)^{-r}\eta^m z^m = (-1)^{r+q} z^m. \tag{2.3}$$

Since σ preserves each $\mathcal{P}_r$ and ρ_a is invariant under $z \mapsto \zeta z$, distinct weight spaces are orthogonal. On $\mathcal{P}_r$, in the basis $z^r, z^{r+d}, \ldots$, the Gram matrix is therefore

$$G^{(r)}_{pq} = (z^{r+dp}, z^{r+dq})_a = (-1)^{r+q} M^{(r)}_{pq}, \qquad M^{(r)}_{pq} = \int_{\mathbb{C}} z^{r+dp}\bar{z}^{r+dq}\rho_a(z)\,\mathrm{d}A(z). \tag{2.4}$$

The substitution $z \mapsto \eta z$ and $\rho_a(\eta z) = \overline{\rho_a(z)}$ give $\overline{M^{(r)}_{qp}} = (-1)^{q-p} M^{(r)}_{pq}$, so each $G^{(r)}$ is a Hermitian matrix.

For $k \geq 1$, let

$$\Delta^{(r)}_k = \det[G^{(r)}_{pq}]_{p,q=0}^{k-1}, \qquad \Delta^{(r)}_0 = 1. \tag{2.5}$$

Assume for the moment that all $\Delta_k^{(r)}$ are nonzero. After the monomials $1, z, \dots, z^N$ are grouped by residue modulo d, their Gram matrix is block diagonal, with leading principal submatrices of the $G^{(r)}$ as blocks. Hence every leading Gram matrix $\big[(z^p, z^q)_a\big]_{p,q=0}^{N}$ is nondegenerate, and Gram–Schmidt gives unique monic orthogonal polynomials $P_n(z) = z^n + \cdots$. Orthogonalizing z^n against lower monomials subtracts only components in the weight space of z^n, so $P_n \in \mathcal{P}_{n \bmod d}$. Following [FH25, Lemma 3.4 and Example 3.5], for $\ell \in \mathbb{Z}$ put

$$\tau(\ell) = \begin{cases} (-1)^{d-1}, & \ell \equiv -1 \pmod d, \\ 1, & \text{otherwise}, \end{cases} \qquad \varrho(n) = (-1)^n \prod_{\ell=0}^{n-1} \tau(\ell). \tag{2.6}$$

If $n = r + dj$ with $0 \le r \le d-1$, exactly j of the integers $0, \dots, n-1$ are congruent to $d-1$, so

$$\varrho(r + dj) = (-1)^{r+dj+(d-1)j} = (-1)^{r+j}. \tag{2.7}$$

The signed norms $\mathfrak{h}_n$ of [FH25, Section 3.1], denoted h_n there, are defined by

$$(P_n, P_n)_a = \varrho(n)\mathfrak{h}_n. \tag{2.8}$$

The matrices $\big[(z^p, z^q)_a\big]_{p,q=0}^{N}$ are exactly the Gram matrices of [FH25, Conjecture 3.6].

3. Gram blocks as Hankel determinants

We write $\binom{k}{2} = k(k-1)/2$ and $\left\{ {q \atop j} \right\}$ for the Stirling numbers of the second kind, defined by $\left\{ {0 \atop 0} \right\} = 1$, $\left\{ {q \atop j} \right\} = 0$ for $j > q$ or $j = 0 < q$, and $\left\{ {q+1 \atop j} \right\} = j\left\{ {q \atop j} \right\} + \left\{ {q \atop j-1} \right\}$. For a function G that is smooth on $(0, \infty)$, induction on q gives

$$\frac{\mathrm{d}^q}{\mathrm{d}t^q} G(\mathrm{e}^t) = \sum_{j=0}^{q} \left\{ {q \atop j} \right\} \mathrm{e}^{jt} G^{(j)}(\mathrm{e}^t), \tag{3.1}$$

because $\frac{\mathrm{d}}{\mathrm{d}t}\big[\mathrm{e}^{jt}G^{(j)}(\mathrm{e}^t)\big] = j\mathrm{e}^{jt}G^{(j)}(\mathrm{e}^t) + \mathrm{e}^{(j+1)t}G^{(j+1)}(\mathrm{e}^t)$.

For $r \in \{0, \dots, d-1\}$ and $\lambda \ge 0$, define the weighted sources

$$\mathcal{W}_r(a; \lambda) = \int_{\mathbb{C}} |z|^{2r} \exp\left(-a|z|^2 + \frac{\mathrm{i}\lambda}{d}(z^d + \bar{z}^d) \right) \mathrm{d}A(z), \qquad F_r(t) = \mathcal{W}_r(a; \mathrm{e}^{t/2}).$$

Lemma 3.1 *(two-source identity).* *For $r \in \{0, \dots, d-1\}$ and all $p, q \ge 0$,*

$$\sum_{k=0}^{p} \sum_{l=0}^{q} \left\{ {p \atop k} \right\} \left\{ {q \atop l} \right\} \left(\frac{\mathrm{i}}{d} \right)^{k+l} M_{kl}^{(r)} = F_r^{(p+q)}(0). \tag{3.2}$$

Proof. For $R > 0$ and $u, v \in \mathbb{C}$, define the disk integral

$$\mathcal{Z}_R(u, v) = \int_{|z| \le R} |z|^{2r} \exp\left(-a|z|^2 + \frac{\mathrm{i}}{d}(uz^d + v\bar{z}^d) \right) \mathrm{d}A(z).$$

Over the compact disk, all derivatives in (u, v) may be taken under the integral sign, and $\mathcal{Z}_R$ is entire in each variable.

Step 1: $\mathcal{Z}_R(u,v) = \mathcal{Z}_R(uv,1)$ for $u,v>0$. For fixed $u,v>0$, the function $H(s) = \mathcal{Z}_R(ue^{s}, ve^{-s})$ is entire in $s\in\mathbb{C}$. The rotation $z\mapsto e^{i\varphi/d}z$, $\varphi\in\mathbb{R}$, preserves the disk, $|z|$ and the area measure, and multiplies z^d by $e^{i\varphi}$ and $\bar z^d$ by $e^{-i\varphi}$; hence $H(s+i\varphi)=H(s)$. Differentiating in φ at $\varphi=0$ gives $iH'(s)=0$, so H is constant, and $H(0)=H(\log v)$ is the claim.

Step 2: the truncated identity. For real t_1,t_2 put $K(t_1,t_2)=\mathcal{Z}_R(e^{t_1},e^{t_2})$. By Step 1, $K(t_1,t_2)=f_R(t_1+t_2)$ with $f_R(t)=\mathcal{Z}_R(e^{t/2},e^{t/2})$, so $\partial_{t_2}^q\partial_{t_1}^p K(0,0) = f_R^{(p+q)}(0)$. On the other hand, (3.1) with $G(w)=\exp(cw)$ gives $\frac{d^p}{dt^p}\exp(ce^t) = \sum_k \left\{{p\atop k}\right\}(ce^t)^k\exp(ce^t)$; applying this under the integral, first with $c=iz^d/d$ and then with $c=i\bar z^d/d$, yields

$$\partial_{t_2}^q\partial_{t_1}^p K(0,0) = \sum_{k=0}^{p}\sum_{l=0}^{q}\left\{{p\atop k}\right\}\left\{{q\atop l}\right\}\left(\frac{i}{d}\right)^{k+l}\int_{|z|\le R} z^{r+dk}\bar z^{r+dl}\rho_a(z)\,dA(z),$$

where $|z|^{2r}z^{dk}\bar z^{dl} = z^{r+dk}\bar z^{r+dl}$.

Step 3: $R\to\infty$. The truncated moments converge to $M_{kl}^{(r)}$, since their integrands are bounded by $|z|^{2r+dk+dl}e^{-a|z|^2}$. For real t, the factor $\exp\big(\frac{i}{d}e^{t/2}(z^d+\bar z^d)\big)$ has modulus 1, and by (3.1) its t-derivatives of order m are this factor times a polynomial in $e^{t/2}(z^d+\bar z^d)$; for $|t|\le 1$ they are bounded by $c_m(1+|z|)^{dm}$. Hence derivatives of all orders may be taken under the integral both for f_R and for F_r, and dominated convergence gives $f_R^{(m)}(0)\to F_r^{(m)}(0)$. Letting $R\to\infty$ in Step 2 proves (3.2). ∎

Corollary 3.2. *For $r\in\{0,\dots,d-1\}$ and $k\ge 1$,*

$$\Delta_k^{(r)} = (-1)^{rk}d^{2\binom{k}{2}}\det\Big[F_r^{(p+q)}(0)\Big]_{p,q=0}^{k-1}. \tag{3.3}$$

Proof. By (2.4), $G_{pq}^{(r)} = (-1)^{r+q}(d/i)^{p+q}\big[(i/d)^{p+q}M_{pq}^{(r)}\big]$. Taking $(d/i)^p$ out of row p and $(-1)^{r+q}(d/i)^q$ out of column q, and using $(d/i)^2=-d^2$, gives $\Delta_k^{(r)} = (-1)^{rk}d^{2\binom{k}{2}}\det[(i/d)^{p+q}M_{pq}^{(r)}]_{p,q<k}$. The left-hand side of (3.2) is the (p,q) entry of $\Sigma\,[(i/d)^{k+l}M_{kl}^{(r)}]\,\Sigma^{\mathsf T}$, where $\Sigma_{pk}=\left\{{p\atop k}\right\}$ is lower triangular with ones on the diagonal; its $k\times k$ truncations therefore have the same determinant. ∎

4. The kernels as Laplace transforms

A random variable G has the Gamma distribution with shape $\beta>0$ if it has density $x^{\beta-1}e^{-x}/\Gamma(\beta)$ on $(0,\infty)$; then $\mathbb{E}[G^{-z}]=\Gamma(\beta-z)/\Gamma(\beta)$ for $\operatorname{Re} z<\beta$. For $r\in\{0,\dots,d-1\}$ let

$$J_r = \{0,1,\dots,d-1\}\setminus\{d-1-r\},\qquad \beta_{r,j}=\frac{r+1+j}{d}\quad (j\in J_r),$$

let $G_{r,j}$, $j\in J_r$, be independent Gamma variables with shapes $\beta_{r,j}$, and put $S_r=\prod_{j\in J_r}G_{r,j}$, a product of $d-1$ factors.

Proposition 4.1 *(Gamma representation).* *For $r\in\{0,\dots,d-1\}$ and $\lambda\ge 0$,*

$$\mathcal{W}_r(a;\lambda) = \pi\, r!\, a^{-r-1}\,\mathbb{E}\left[\exp\big(-\lambda^2 d^{\,d-2}a^{-d}S_r\big)\right]. \tag{4.1}$$

In particular, $\int_{\mathbb{C}} |z|^{2r} \rho_a \, \mathrm{d}A = \pi r!\, a^{-r-1} \mathbb{E}[\exp(-d^{\,d-2} a^{-d} S_r)]$, *and* $F_r(t) = \pi r!\, a^{-r-1} \mathbb{E}[\exp(-\mathrm{e}^t \kappa S_r)]$ *with* $\kappa = d^{\,d-2} a^{-d}$.

Proof. At $\lambda = 0$ both sides equal $\pi r!\, a^{-r-1}$. The substitution $z \mapsto a^{-1/2} z$ gives $\mathcal{W}_r(a;\lambda) = a^{-r-1} \mathcal{W}_r(1; \lambda a^{-d/2})$, so we may take $a = 1$ and $\lambda > 0$.

Step 1: a radial integral. In polar coordinates $z = R\mathrm{e}^{\mathrm{i}\theta}$, $z^d + \bar{z}^d = 2R^d \cos(d\theta)$, and $\int_0^{2\pi} \mathrm{e}^{\mathrm{i}x\cos(d\theta)} \, \mathrm{d}\theta = 2\pi J_0(x)$ [OLBC10, equation (10.9.2)]. With $u = R^2$,

$$\mathcal{W}_r(1;\lambda) = \pi \int_0^\infty u^r \mathrm{e}^{-u} J_0\big(c\lambda u^{d/2}\big) \, \mathrm{d}u, \qquad c = \frac{2}{d}.$$

Step 2: the Mellin transform. Write $g(x) = \mathcal{W}_r(1;\sqrt{x})$ for $x > 0$, and let $0 < \operatorname{Re} z = \sigma < \min(\frac{1}{4}, \frac{r+1}{d})$. Since $|J_0(y)| \le 1$ and $|J_0(y)| \le C y^{-1/2}$ [OLBC10, equations (10.14.1) and (10.17.3)], $\int_0^\infty y^{2\sigma-1} |J_0(y)| \, \mathrm{d}y < \infty$. The substitution $y = c\sqrt{x}\, u^{d/2}$ gives $\int_0^\infty x^{\sigma-1} |J_0(c\sqrt{x}\, u^{d/2})| \, \mathrm{d}x = 2(cu^{d/2})^{-2\sigma} \int_0^\infty y^{2\sigma-1} |J_0(y)| \, \mathrm{d}y$, and $\int_0^\infty u^{r-d\sigma} \mathrm{e}^{-u} \, \mathrm{d}u < \infty$. Hence Fubini's theorem applies, and with $\int_0^\infty y^{2z-1} J_0(y) \, \mathrm{d}y = 2^{2z-1}\Gamma(z)/\Gamma(1-z)$ [OLBC10, equation (10.22.43)] we obtain

$$\int_0^\infty x^{z-1} g(x) \, \mathrm{d}x = \pi \, d^{2z} \, \frac{\Gamma(z)\, \Gamma(r+1-dz)}{\Gamma(1-z)}.$$

Step 3: Gauss's multiplication formula. By [OLBC10, equation (5.5.6)], applied at $r+1-dz$ and at $r+1$,

$$\frac{\Gamma(r+1-dz)}{\Gamma(r+1)} = d^{-dz} \prod_{j=0}^{d-1} \frac{\Gamma\big(\frac{r+1+j}{d} - z\big)}{\Gamma\big(\frac{r+1+j}{d}\big)}.$$

The factor with $j = d-1-r$ equals $\Gamma(1-z)$. Therefore

$$\int_0^\infty x^{z-1} g(x) \, \mathrm{d}x = \pi \, r! \, \Gamma(z) \, \big(d^{\,d-2}\big)^{-z} \prod_{j \in J_r} \frac{\Gamma(\beta_{r,j} - z)}{\Gamma(\beta_{r,j})} = \pi \, r! \, \Gamma(z) \, \mathbb{E}\big[(d^{\,d-2} S_r)^{-z}\big].$$

Step 4: identification. Let $\tilde{g}(x) = \pi r! \, \mathbb{E}[\exp(-x\, d^{\,d-2} S_r)]$. By Tonelli's theorem and $\int_0^\infty x^{z-1} \mathrm{e}^{-xs} \, \mathrm{d}x = \Gamma(z) s^{-z}$, the Mellin transform of $\tilde{g}$ is the last expression for $0 < \operatorname{Re} z < \min_{j \in J_r} \beta_{r,j}$, a strip that contains the line $\operatorname{Re} z = \sigma$ of Step 2. Both g and $\tilde{g}$ are continuous on $(0,\infty)$, and $x^{\sigma-1} g$, $x^{\sigma-1} \tilde{g}$ are integrable. Under $x = \mathrm{e}^s$, their Mellin transforms on $\operatorname{Re} z = \sigma$ are the Fourier transforms of the integrable functions $\mathrm{e}^{\sigma s} g(\mathrm{e}^s)$ and $\mathrm{e}^{\sigma s} \tilde{g}(\mathrm{e}^s)$; these coincide, so $g = \tilde{g}$. ∎

5. Signs of the Hankel determinants

For real $x_1, \dots, x_k$ write $V(x_1, \dots, x_k) = \prod_{1 \le i < j \le k} (x_j - x_i)$.

Lemma 5.1 *(Hankel signs).* *Let* $m \ge 1$, $\beta_1, \dots, \beta_m > 0$ *and* $\kappa > 0$, *let* $G_1, \dots, G_m$ *be independent Gamma variables with shapes* $\beta_1, \dots, \beta_m$, *and let* $F(t) = \mathbb{E}[\exp(-\mathrm{e}^t \kappa G_1 \cdots G_m)]$. *Then, for every real* t *and every* $k \ge 1$,

$$(-1)^{\binom{k}{2}} \det\Big[F^{(p+q)}(t)\Big]_{p,q=0}^{k-1} > 0.$$

Proof. The variable $\log G_j$ has the density $q_j(v) = \mathrm{e}^{\beta_j v - \mathrm{e}^v}/\Gamma(\beta_j)$ on $\mathbb{R}$. Put $\phi_0(u) = \exp(-\mathrm{e}^u)$ and, for $j = 1, \dots, m$,

$$\phi_j(u) = \mathbb{E}\big[\phi_{j-1}(u + \log G_j)\big] = \int_{\mathbb{R}} \phi_{j-1}(w)\, q_j(w-u)\, \mathrm{d}w,$$

so that $F(t) = \phi_m(t + \log\kappa)$. By (3.1) with $G(x) = \mathrm{e}^{-x}$, $\phi_0^{(p)}(u) = \mathrm{e}^{-x}\sum_i \left\{ {p \atop i} \right\}(-x)^i$ with $x = \mathrm{e}^u$; in particular $\phi_0^{(p)}$ is bounded. By induction and dominated convergence, each ϕ_j is smooth with bounded derivatives $\phi_j^{(p)}(u) = \mathbb{E}[\phi_{j-1}^{(p)}(u + \log G_j)] = \int \phi_{j-1}^{(p)}(w) q_j(w-u)\,\mathrm{d}w$.

Claim: for $0 \le j \le m-1$ *and* $u_1 < \cdots < u_k$, $\operatorname{sgn}\det[\phi_j^{(p)}(u_l)]_{p<k,\, l\le k} = (-1)^{\binom{k}{2}}$. For $j = 0$, the matrix Σ of Corollary 3.2 is unitriangular, so with $x_l = \mathrm{e}^{u_l}$,

$$\det[\phi_0^{(p)}(u_l)] = \mathrm{e}^{-\sum_l x_l}\, V(-x_1, \dots, -x_k) = (-1)^{\binom{k}{2}} \mathrm{e}^{-\sum_l x_l}\, V(x_1, \dots, x_k),$$

and $V(x) > 0$ because the x_l increase. For the induction step, Andréief's identity [And86, For19], applied to the bounded functions $\phi_{j-1}^{(p)}$ and the integrable functions $w \mapsto q_j(w - u_l)$, together with the symmetry of the integrand, gives

$$\det[\phi_j^{(p)}(u_l)] = \int_{w_1 < \cdots < w_k} \det[\phi_{j-1}^{(p)}(w_i)]_{p,i}\, \det[q_j(w_i - u_l)]_{i,l} \prod_i \mathrm{d}w_i.$$

Here $q_j(w-u) = \Gamma(\beta_j)^{-1}\mathrm{e}^{\beta_j w}\mathrm{e}^{-\beta_j u}\exp(\xi\zeta)$ with $\xi = \mathrm{e}^w$ and $\zeta = -\mathrm{e}^{-u}$, both strictly increasing. The kernel $\exp(\xi\zeta)$ is strictly totally positive: for $\xi_1 < \cdots < \xi_k$ and $\zeta_1 < \cdots < \zeta_k$, $\det[\mathrm{e}^{\xi_i\zeta_l}] > 0$. Indeed, a nonzero sum $\sum_l c_l \mathrm{e}^{\zeta_l \xi}$ has at most $k-1$ real zeros, by induction on k with Rolle's theorem applied to $\frac{\mathrm{d}}{\mathrm{d}\xi}(\mathrm{e}^{-\zeta_1\xi}\sum_l c_l \mathrm{e}^{\zeta_l\xi})$; so the determinant never vanishes on the connected set of ordered (ξ, ζ). To find its sign, fix $x_1 < \cdots < x_k$ and put $\xi_i = \varepsilon x_i$. Expanding $\mathrm{e}^{\varepsilon x_i \zeta_l} = \sum_{p\ge 0}(\varepsilon x_i \zeta_l)^p/p!$ and applying the Cauchy–Binet formula gives

$$\det[\mathrm{e}^{\varepsilon x_i \zeta_l}] = \varepsilon^{\binom{k}{2}} \frac{V(x_1, \dots, x_k)\, V(\zeta_1, \dots, \zeta_k)}{\prod_{p=0}^{k-1} p!} + O\big(\varepsilon^{\binom{k}{2}+1}\big) \qquad (\varepsilon \downarrow 0),$$

which is positive for small ε. Hence the second determinant is positive on the ordered region, the first has sign $(-1)^{\binom{k}{2}}$ by induction, and the claim follows.

Now let $u = t + \log\kappa$ and $H_{pq} = \phi_m^{(p+q)}(u)$. From $q_m' = q_m \cdot (\beta_m - \mathrm{e}^v)$, induction gives $q_m^{(q)}(v) = q_m(v) P_q(\mathrm{e}^v)$, where $P_0 = 1$ and $P_{q+1}(y) = (\beta_m - y)P_q(y) + yP_q'(y)$, so that P_q has degree q and leading coefficient $(-1)^q$. On $|u| \le M$, $|q_m^{(q)}(w-u)| \le C_{q,M}\mathrm{e}^{\beta_m w}(1 + \mathrm{e}^w)^q \exp(-\mathrm{e}^{-M}\mathrm{e}^w)$, which is integrable in w. Differentiating $\phi_m^{(p)}(u) = \int \phi_{m-1}^{(p)}(w) q_m(w - u)\,\mathrm{d}w$ a further q times in u therefore gives

$$H_{pq} = \int_{\mathbb{R}} \phi_{m-1}^{(p)}(w)\, g_q(w)\,\mathrm{d}w, \qquad g_q(w) = (-1)^q q_m^{(q)}(w-u) = q_m(w-u)\,\widetilde{P}_q(\mathrm{e}^{w-u}),$$

with $\widetilde{P}_q = (-1)^q P_q$ monic of degree q. Andréief's identity gives

$$\det H = \int_{w_1 < \cdots < w_k} \det[\phi_{m-1}^{(p)}(w_i)] \prod_i q_m(w_i - u)\; V\big(\mathrm{e}^{w_1 - u}, \dots, \mathrm{e}^{w_k - u}\big) \prod_i \mathrm{d}w_i,$$

because $\det[\widetilde{P}_q(y_i)] = V(y)$ for monic $\widetilde{P}_q$. On the ordered region $V(\mathrm{e}^{w_i - u}) > 0$, and by the claim the integrand has sign $(-1)^{\binom{k}{2}}$ and does not vanish. Hence $(-1)^{\binom{k}{2}} \det H > 0$. ∎

Proposition 5.2 *(block determinants).* *For $r \in \{0, \dots, d-1\}$ and $k \geq 1$, $\operatorname{sgn} \Delta_k^{(r)} = (-1)^{rk+\binom{k}{2}}$.*

Proof. By Proposition 4.1, $F_r(t) = \pi r!\, a^{-r-1} F(t)$, where F is the function of Lemma 5.1 for the $m = d-1 \geq 1$ shapes $\beta_{r,j}$, $j \in J_r$, and $\kappa = d^{\,d-2} a^{-d}$. Hence $\operatorname{sgn} \det[F_r^{(p+q)}(0)] = (-1)^{\binom{k}{2}}$, and Corollary 3.2 gives the claim. ∎

Proof of Theorem 1.1(i). By Proposition 5.2, every $\Delta_k^{(r)}$ is nonzero, so by Section 2 every leading Gram matrix is nondegenerate and the polynomials P_n exist. Let $n = r + dj$. Replacing z^n by P_n in the basis $z^r, z^{r+d}, \dots, z^{r+dj}$ is a unitriangular change of basis, and orthogonality makes the new Gram matrix $\operatorname{diag}\big([G_{pq}^{(r)}]_{p,q=0}^{j-1}, (P_n, P_n)_a\big)$. Comparing determinants gives

$$(P_n, P_n)_a = \frac{\Delta_{j+1}^{(r)}}{\Delta_j^{(r)}}; \tag{5.1}$$

this Schur-complement identity uses only the nonvanishing of the leading minors, not positive definiteness. By Proposition 5.2 and $\binom{j+1}{2} - \binom{j}{2} = j$, $\operatorname{sgn}(P_n, P_n)_a = (-1)^{r+j}$, which is $\varrho(n)$ by (2.7). Hence $\mathfrak{h}_n = \varrho(n)(P_n, P_n)_a > 0$. The last assertion is the identification at the end of Section 2. ∎

Remark 5.3. Unwinding the proof of Lemma 5.1 writes $(-1)^{rk+\binom{k}{2}} \Delta_k^{(r)}$ as a $k(d-1)$-fold integral of a positive function. By (5.1), the norms $\mathfrak{h}_n$ and the ratios V_n can therefore be evaluated without the cancellation inherent in the oscillatory moment determinants.

6. The recurrence

Let $d \geq 3$. Define $V_n = \mathfrak{h}_{n+1}/\mathfrak{h}_n$ for $n \geq 0$ and $V_j = 0$ for $j < 0$. By Theorem 1.1(i), $V_n > 0$ for every $n \geq 0$. We derive (1.1), following [FH25, Lemma 3.4 and equations (3.1)–(3.3)].

If g has weight ℓ, then

$$\sigma(zg) = -\tau(\ell) z\, \sigma g. \tag{6.1}$$

For $\ell \leq d-2$ this follows from $(-\eta)^{-\ell-1}\eta = -(-\eta)^{-\ell}$; for $\ell = d-1$ the weight returns to 0, and $\sigma(zg) = \eta z\, g(\eta z) = \eta(-\eta)^{d-1} z\, \sigma g = (-1)^{d-1}\eta^d z\, \sigma g = -\tau(d-1) z\, \sigma g$.

Let $L = \partial_z + \mathrm{i} z^{d-1}$, where $\partial_z = \frac{1}{2}(\partial_{\operatorname{Re} z} - \mathrm{i}\partial_{\operatorname{Im} z})$. Since $\partial_z \overline{\sigma g(z)} = 0$ and $\partial_z \rho_a(z) = (-a\bar{z} + \mathrm{i} z^{d-1})\rho_a(z)$, integration by parts in (2.2), with no boundary terms because of the Gaussian decay, and (6.1) give, for g of weight ℓ,

$$(Lf, g)_a = a \int_{\mathbb{C}} f\, \bar{z}\, \overline{\sigma g}\, \rho_a \, \mathrm{d}A = -\tau(\ell) a (f, zg)_a. \tag{6.2}$$

The polynomials zP_n and LP_n have weights $n+1$ and $n-1$ modulo d, degrees $n+1$ and $n+d-1$, and leading coefficients 1 and i. If $j \leq n+1-2d$ and $j \equiv n+1 \pmod d$, then $\deg LP_j < n$, so (6.2) gives $0 = (LP_j, P_n)_a = -\tau(n) a (P_j, zP_n)_a$. If $j \leq n-1-d$ and $j \equiv n-1 \pmod d$, then $\deg zP_j < n$, so $(LP_n, P_j)_a = -\tau(j) a (P_n, zP_j)_a = 0$. Hence

$$zP_n = P_{n+1} + c_n P_{n+1-d}, \qquad LP_n = \mathrm{i} P_{n+d-1} + d_n P_{n-1}, \tag{6.3}$$

where terms with negative indices are omitted; equivalently, $c_n = 0$ for $n \le d-2$, and c_n for $n \ge d-1$ and d_n for $n \ge 1$ are the coefficients displayed. The $d-1$ consecutive integers $n-d+1, \dots, n-1$ miss exactly the residue of n modulo d, and the product of τ over a full residue system is $(-1)^{d-1}$; hence (2.6) gives

$$\varrho(n) = \tau(n)\varrho(n-d+1) \quad (n \ge d-1), \qquad \varrho(n) = -\tau(n-1)\varrho(n-1) \quad (n \ge 1).$$

Pairing (6.3) with P_{n+1-d} (for $n \ge d-1$) and P_{n-1} (for $n \ge 1$), and using (6.2) and Hermitian symmetry, we obtain

$$\begin{aligned} \mathrm{i}\varrho(n)\mathfrak{h}_n &= (LP_{n+1-d}, P_n)_a = -\tau(n)a(P_{n+1-d}, zP_n)_a, \\ c_n &= \frac{(zP_n, P_{n+1-d})_a}{\varrho(n+1-d)\mathfrak{h}_{n+1-d}} = \frac{\mathrm{i}}{a}\frac{\mathfrak{h}_n}{\mathfrak{h}_{n+1-d}}, \\ d_n &= \frac{(LP_n, P_{n-1})_a}{\varrho(n-1)\mathfrak{h}_{n-1}} = -\tau(n-1)a\frac{(P_n, zP_{n-1})_a}{\varrho(n-1)\mathfrak{h}_{n-1}} = a\frac{\mathfrak{h}_n}{\mathfrak{h}_{n-1}}; \end{aligned}$$

the first line holds because the leading term of LP_{n+1-d} is $\mathrm{i}P_n$, and the last because that of zP_{n-1} is P_n. Thus $c_m = \frac{\mathrm{i}}{a}\prod_{i=m+1-d}^{m-1} V_i$ for every $m \ge 0$, including $c_m = 0$ for $m \le d-2$.

Applying the first relation in (6.3) $d-1$ times to P_n, a path from P_n to P_{n-1} in $d-1$ steps of $+1$ or $-(d-1)$ uses exactly one step of $-(d-1)$. So the coefficient of P_{n-1} in $z^{d-1}P_n$ is $\sum_{s=0}^{d-2} c_{n+s}$. Let $n \ge 1$. The coefficient of P_{n-1} in $\partial_z P_n$ is n, since $\partial_z P_n = nz^{n-1} + \cdots$ lies in the weight space of P_{n-1}. Comparing coefficients of P_{n-1} in $LP_n = \partial_z P_n + \mathrm{i}z^{d-1}P_n$ and using $d_n = aV_{n-1}$, we obtain, with $m = n-1 \ge 0$,

$$aV_m = m + 1 - \frac{1}{a}\sum_{s=0}^{d-2}\prod_{i=m+s-d+2}^{m+s} V_i,$$

which is (1.1).

Proof of Theorem 1.1(ii). The sequence $V_n = \mathfrak{h}_{n+1}/\mathfrak{h}_n$ is positive and satisfies (1.1). For $j \le d-1$, $P_j = z^j$, $\sigma z^j = (-1)^j z^j$ by (2.3), and $\varrho(j) = (-1)^j$; hence $\mathfrak{h}_j = \int_{\mathbb{C}} |z|^{2j}\rho_a \,\mathrm{d}A$, and V_j is given by (1.2) for $j \le d-3$. If $(\widetilde{V}_n)$ is another real solution with these initial values and $\widetilde{V}_i = V_i$ for $i \le m+d-3$, then (1.1) at index m is linear in $\widetilde{V}_{m+d-2}$ with coefficient $a^{-2}\widetilde{V}_m \cdots \widetilde{V}_{m+d-3} = a^{-2}V_m \cdots V_{m+d-3} > 0$, so $\widetilde{V}_{m+d-2} = V_{m+d-2}$. Induction proves uniqueness. ∎

7. The quartic case

In this section $d = 4$. Let $y > 0$, $\alpha \ge 0$, $\beta \ge 0$, and $b = \beta + \frac{1}{2}$. For $w > 0$, define

$$\Phi_{\alpha,\beta}(w; y) = \int_0^\infty \mathrm{e}^{-ys} s^{\alpha-1/2}(1+ws^2)^{-b}\,\mathrm{d}s. \tag{7.1}$$

Throughout, $y = a^2/2$ and $C = \sqrt{\pi/2}$.

Lemma 7.1 *(quartic kernels).* *Define*

r	A_r	α_r	β_r
0	C	0	0
1	aC	1	0
2	$2C$	1	1

(7.2)

Then $\mathcal{W}_r(a;\lambda) = A_r\Phi_{\alpha_r,\beta_r}(\lambda^2;y)$ *for* $r = 0,1,2$ *and* $\lambda > 0$.

Proof. Let $\nu = \alpha + \frac{1}{2}$. Since $(1+ws^2)^{-b} = \mathbb{E}[\mathrm{e}^{-ws^2G_b}]$ for a Gamma variable G_b with shape b, and $\mathrm{e}^{-ys}s^{\nu-1}$ is $\Gamma(\nu)y^{-\nu}$ times the density of G_ν/y,

$$\Phi_{\alpha,\beta}(w;y) = \Gamma(\nu)y^{-\nu}\,\mathbb{E}\left[\exp\left(-wy^{-2}G_\nu^2G_b\right)\right]$$

with G_ν and G_b independent. By the duplication formula [OLBC10, equation (5.5.5)], G_ν^2 has the same Mellin transform, and hence the same law, as $4G_{\nu/2}G_{(\nu+1)/2}$ with independent factors. So $\Phi_{\alpha,\beta}(w;y) = \Gamma(\nu)y^{-\nu}\mathbb{E}[\exp(-4wy^{-2}G_{\nu/2}G_{(\nu+1)/2}G_b)]$. For $(\alpha,\beta) = (\alpha_r,\beta_r)$ the shapes $\{\nu/2,(\nu+1)/2,b\}$ are $\{\frac14,\frac34,\frac12\}$, $\{\frac34,\frac54,\frac12\}$ and $\{\frac34,\frac54,\frac32\}$, which are the shapes $\beta_{r,j}$, $j\in J_r$, of Section 4 for $r=0,1,2$. Moreover $4y^{-2} = 16a^{-4} = d^{\,d-2}a^{-d}$ and $A_r\Gamma(\nu)y^{-\nu} = \pi r!\,a^{-r-1}$. The claim follows from Proposition 4.1. ∎

Lemma 7.2 *(Bessel form).* *For every* $a>0$, $\int_{\mathbb{C}}\rho_a\,\mathrm{d}A = h(a)$, *the Bessel combination* (1.5).

Proof. Put $H(a) = \int_{\mathbb{C}}\rho_a\,\mathrm{d}A$ and $m_{pq} = \int_{\mathbb{C}} z^p\bar z^q\rho_a(z)\,\mathrm{d}A(z)$ for $a>0$.

Step 1: $H'''+a^2H'+aH = 0$ *on* $(0,\infty)$. Differentiation under the integral sign gives $H^{(k)}(a) = (-1)^k m_{kk}$. With the Wirtinger derivative $\partial_z = \frac12(\partial_{\mathrm{Re}\,z} - \mathrm{i}\partial_{\mathrm{Im}\,z})$, $\partial_z\bar z^q = 0$ and $\partial_z\rho_a = (-a\bar z + \mathrm{i}z^3)\rho_a$, so integrating $\partial_z(z^p\bar z^q\rho_a)$ over $\mathbb{C}$, with no boundary terms because of the Gaussian decay, gives

$$p\,m_{p-1,q} + \mathrm{i}\,m_{p+3,q} = a\,m_{p,q+1}.$$

For $(p,q) = (1,0)$ and $(0,3)$ this reads $m_{00}+\mathrm{i}m_{40} = am_{11}$ and $\mathrm{i}m_{33} = am_{04}$. The substitution $z\mapsto\eta z$ of Section 2 gives $\overline{m_{qp}} = \bar\eta^{\,p}\eta^q m_{pq}$; hence every m_{kk} is real and $\overline{m_{40}} = -m_{04}$. Therefore $m_{33} = am_{00} - a^2m_{11}$, which is the claimed equation.

Step 2: the Bessel side. For $x>0$, $J_\nu(x) = (x/2)^\nu s_\nu(x)$ with the even entire function $s_\nu(x) = \sum_{k\ge0}(-1)^kx^{2k}/\big(4^kk!\,\Gamma(k+\nu+1)\big)$ [OLBC10, equation (10.2.2)], and Bessel's equation [OLBC10, equation (10.2.1)] becomes $xs_\nu'' + (2\nu+1)s_\nu' + xs_\nu = 0$. Writing $s_\pm = s_{\pm1/4}$ and $x = a^2/4$, so that $(x/2)^{\mp1/4} = (a^2/8)^{\mp1/4}$, we get, for $a>0$,

$$h(a) = h_S(a) := \frac{\pi^2}{4}\left[\sqrt8\,s_-(x)^2 - \sqrt2\,a\,s_-(x)s_+(x) + \frac{a^2}{\sqrt8}\,s_+(x)^2\right].$$

The function h_S is smooth on $\mathbb{R}$. Put $U(a) = s_-(a^2/4)$ and $V(a) = a\,s_+(a^2/4)$. The equations for $s_\mp$, with $2\nu+1 = \frac12$ and $\frac32$, give $U''+\frac{a^2}{4}U = 0$ and $V''+\frac{a^2}{4}V = 0$. If y_1,y_2 solve $y''+qy=0$, then $P = y_1y_2$ satisfies $P'''+4qP'+2q'P = 0$; with $q = a^2/4$ this

is $P''' + a^2P' + aP = 0$. Since $h_S = \frac{\pi^2}{4}\big(\sqrt{8}\,U^2 - \sqrt{2}\,UV + V^2/\sqrt{8}\big)$, we get $h_S''' + a^2h_S' + ah_S = 0$ on $\mathbb{R}$. Since $s_\nu(0) = 1/\Gamma(\nu+1)$ and $s_\nu'(0) = 0$,

$$h_S(0) = \frac{\pi^2\sqrt{8}}{4\,\Gamma(\frac34)^2}, \qquad h_S'(0) = -\frac{\pi^2\sqrt{2}}{4\,\Gamma(\frac34)\Gamma(\frac54)}, \qquad h_S''(0) = \frac{\pi^2}{2\sqrt{8}\,\Gamma(\frac54)^2}.$$

Step 3: the planar side as $a \to 0^+$. By Lemma 7.1 with $\lambda = 1$,

$$\begin{aligned} H &= C\,\Phi_{0,0}(1;y), \\ H' &= -\mathcal{W}_1(a;1) = -aC\,\Phi_{1,0}(1;y), \\ H'' &= \mathcal{W}_2(a;1) = 2C\,\Phi_{1,1}(1;y). \end{aligned}$$

As $y \downarrow 0$, monotone convergence and $\int_0^\infty s^{2x-1}(1+s^2)^{-x-z}\,\mathrm{d}s = \frac12 B(x,z) = \frac{\Gamma(x)\Gamma(z)}{2\Gamma(x+z)}$ [OLBC10, equations (5.12.1) and (5.12.3)] give $\Phi_{0,0}(1;y) \to \frac12 B(\frac14,\frac14) = \Gamma(\frac14)^2/(2\sqrt{\pi})$ and $\Phi_{1,1}(1;y) \to \frac12 B(\frac34,\frac34) = \Gamma(\frac34)^2/\sqrt{\pi}$. The substitution $s = u/y$ gives $a\Phi_{1,0}(1;y) = \sqrt{2}\int_0^\infty \mathrm{e}^{-u}u^{1/2}(y^2+u^2)^{-1/2}\,\mathrm{d}u$; the integrand is bounded by $\mathrm{e}^{-u}u^{-1/2}$, so by dominated convergence this tends to $\sqrt{2}\,\Gamma(\frac12) = \sqrt{2\pi}$. Using $\Gamma(\frac14)\Gamma(\frac34) = \pi\sqrt{2}$, $\Gamma(\frac54) = \frac14\Gamma(\frac14)$ and $\Gamma(\frac12) = \sqrt{\pi}$ [OLBC10, equations (5.5.3), (5.5.1), (5.4.6)], the limits of H, H', H'' as $a \to 0^+$ are $h_S(0)$, $h_S'(0) = -\pi$ and $h_S''(0)$.

Step 4: uniqueness. The function $W = H - h_S$ is smooth on $(0,\infty)$, satisfies $W''' = -a^2W' - aW$, and $W, W', W'' \to 0$ as $a \to 0^+$. Fix $a_1 > 0$. For $0 < a \le a_1$, the energy $E = W^2 + W'^2 + W''^2$ satisfies $E' \le (2 + a_1^2 + a_1)E$, by $2|uv| \le u^2 + v^2$. Hence $\mathrm{e}^{-(2+a_1^2+a_1)a}E(a)$ is nonincreasing on $(0,a_1]$ and tends to 0 as $a \to 0^+$, so $E = 0$ there. Thus $H = h_S = h$ on $(0,\infty)$. ∎

The identity of Lemma 7.2 also follows from Lemma 7.1 and the Dixon–Ferrar formula [GR15, entry 6.518]: substitute $s = \sinh t$ in $\Phi_{0,0}(1;y)$, use $K_{1/2}(z) = \sqrt{\pi/(2z)}\,\mathrm{e}^{-z}$, take $\nu = \frac14$ and $x = a^2/4$ in the formula, and use $Y_{1/4} = J_{1/4} - \sqrt{2}\,J_{-1/4}$.

Proof of Theorem 1.2. With $d = 4$, $V_m = av_m$ and $\epsilon = a^{-2}$, (1.1) is (1.3). By Lemma 7.2, $h(a) = \int_{\mathbb{C}} \rho_a\,\mathrm{d}A$, and differentiation under this absolutely convergent integral gives $h'(a) = -\int_{\mathbb{C}} |z|^2\rho_a\,\mathrm{d}A$ and $h''(a) = \int_{\mathbb{C}} |z|^4\rho_a\,\mathrm{d}A$. By the proof of Theorem 1.1(ii),

$$\mathfrak{h}_0 = h, \qquad \mathfrak{h}_1 = -h', \qquad \mathfrak{h}_2 = h''.$$

Hence $h(a) > 0$ and $h'(a) < 0$, so (1.4) is well defined, and $v_n = \mathfrak{h}_{n+1}/(a\mathfrak{h}_n)$ reproduces it. Theorem 1.1(ii) for $d = 4$ now gives existence, uniqueness and positivity. (For $d = 4$ the initial data of (1.2) consist of V_0 and V_1, that is, of v_0 and v_1.) ∎

8. Uniqueness of the positive solution

Throughout this section $d \ge 3$ and $\epsilon > 0$ are fixed. For a positive sequence $(v_n)_{n\ge0}$, with $v_k = 0$ for $k < 0$, put $T_m = v_m v_{m+1}\cdots v_{m+d-2}$ for $m \ge 0$ and $T_m = 0$ for $m < 0$; then (1.6) reads

$$v_n + \sum_{m=n-d+2}^{n} T_m = (n+1)\epsilon, \qquad n \ge 0. \tag{8.1}$$

For $x, y > 0$, the logarithmic mean is $\mathrm{LM}(x,y) = (y-x)/(\log y - \log x)$ if $x \neq y$ and $\mathrm{LM}(x,x) = x$; it lies between x and y, and

$$\mathrm{LM}(x,y) \;\geq\; \frac{x}{1+|\log(y/x)|}. \tag{8.2}$$

Indeed, this is clear if $y \geq x$; if $y = x\mathrm{e}^{-s}$ with $s > 0$, then $\mathrm{LM}(x,y) = x(1-\mathrm{e}^{-s})/s \geq x/(1+s)$ because $\mathrm{e}^s \geq 1+s$. For finitely supported real h and real z we use the identity

$$\Big(\sum_i z_i\Big)\Big(\sum_i h_i^2 z_i\Big) \;=\; \Big(\sum_i h_i z_i\Big)^2 + \sum_{i<j}(h_i - h_j)^2 z_i z_j. \tag{8.3}$$

Lemma 8.1 *(crude bounds).* *Put $K = 1 + (d-1)^{d-1}\epsilon^{d-2}$. Every positive solution of (8.1) satisfies*

$$\frac{\epsilon}{K}(n+1)^{-(d-3)} \;\leq\; v_n \;\leq\; (n+1)\epsilon, \qquad T_n \;\leq\; (n+1)\epsilon \qquad (n \geq 0).$$

Proof. All terms in (8.1) are nonnegative, which gives the upper bounds. Factoring v_n out of (8.1) leaves the bracket $1 + \sum\prod v_i$, a sum over the $d-1$ windows containing n of the products of their other $d-2$ entries. Each such entry has $i \leq n+d-2$, so $v_i \leq (i+1)\epsilon \leq (d-1)(n+1)\epsilon$, and the bracket is at most $1 + (d-1)\big((d-1)(n+1)\epsilon\big)^{d-2} \leq K(n+1)^{d-2}$. ∎

An approximate solution. For $t > 0$, let $q(t) > 0$ be the root of $q + (d-1)q^{d-1} = \epsilon t$; since the left-hand side is increasing in q, q is well defined, smooth and increasing. Put $q_n = q(n+1)$, $Q_m = q_m q_{m+1} \cdots q_{m+d-2}$ for $m \geq 0$, $Q_m = 0$ for $m < 0$, and

$$r_n \;=\; q_n + \sum_{m=n-d+2}^{n} Q_m - (n+1)\epsilon.$$

Lemma 8.2 *(the approximate solution).* *For $t \geq 1$, $q(1)t^{1/(d-1)} \leq q(t) \leq \big(\epsilon t/(d-1)\big)^{1/(d-1)}$ and $q(1) \geq \epsilon/K$. Moreover $Q_m \leq (m+1)\epsilon$ and $|r_n| \leq 4\epsilon(d-1)^4/(n+1)$ for all $m, n \geq 0$.*

Proof. The upper bound follows from $(d-1)q^{d-1} \leq \epsilon t$. For $x = q(1)t^{1/(d-1)}$ and $t \geq 1$, $x + (d-1)x^{d-1} \leq t\big(q(1) + (d-1)q(1)^{d-1}\big) = \epsilon t$, so $q(t) \geq x$. For $x = \epsilon/K$, $x + (d-1)x^{d-1} = \frac{\epsilon}{K}\big(1 + (d-1)\epsilon^{d-2}K^{-(d-2)}\big) \leq \epsilon$, so $q(1) \geq \epsilon/K$. By the upper bound, $Q_m \leq \epsilon(m+d-1)/(d-1) \leq (m+1)\epsilon$.

Let $f = \log q$. Differentiating $q + (d-1)q^{d-1} = \epsilon t$ gives $f' = \epsilon/\big(q + (d-1)^2 q^{d-1}\big)$, so $0 < f'(t) \leq 1/t$, and

$$f'' \;=\; -f'^2\,\frac{1 + (d-1)^3 q^{d-2}}{1 + (d-1)^2 q^{d-2}}, \qquad |f''(t)| \;\leq\; \frac{d-1}{t^2}.$$

Let $t = n+1 \geq 2(d-1)$. For $0 \leq j \leq d-2$ and $0 \leq x \leq 1$ put $P_j(x) = \prod_{k=0}^{d-2} q\big(t + x(k-j)\big)$; all arguments lie in $[t/2, 3t/2]$. Since $n \geq d-2$, $\sum_{m=n-d+2}^{n} Q_m = \sum_j Q_{n-j} = \sum_j P_j(1)$, while $\sum_j P_j(0) = (d-1)q(t)^{d-1} = \epsilon t - q(t)$ and $\sum_j P_j'(0) = q(t)^{d-1} f'(t) \sum_{j,k}(k -$

$j) = 0$. Hence $r_n = \sum_j \big(P_j(1) - P_j(0) - P_j'(0)\big)$. On $[t/2, 3t/2]$ we have $f' \le 2/t$ and $|f''| \le 4(d-1)/t^2$, so $|(\log P_j)'| \le (d-1)^2/t$ and $|(\log P_j)''| \le 4(d-1)^4/t^2$; also $P_j \le q(3t/2)^{d-1} \le 3\epsilon t/(2(d-1))$. Therefore $|P_j''| = P_j\,|(\log P_j)'' + ((\log P_j)')^2| \le 15\epsilon(d-1)^3/(2t)$, and Taylor's formula gives $|r_n| \le \frac{1}{2}(d-1)\cdot 15\epsilon(d-1)^3/(2t) \le 4\epsilon(d-1)^4/t$. If $t < 2(d-1)$, then $-\epsilon t \le r_n \le \sum_m Q_m \le (d-1)\epsilon t$, and $(d-1)\epsilon t < 4\epsilon(d-1)^3/t$. ∎

Proposition 8.3 *(a priori bound).* *There is a constant M, depending only on d and ϵ, such that every positive solution of (8.1) satisfies*

$$\sum_{n\ge 0} \log^2 \frac{v_n}{q_n} \le M^2.$$

In particular $v_n/q_n \to 1$, and $v_n\big((d-1)/(\epsilon(n+1))\big)^{1/(d-1)} \to 1$.

Proof. Put $\delta_n = \log(v_n/q_n)$ and $H(t) = \log K + (d-2)\log t$. By Lemmas 8.1 and 8.2, both v_n and q_n lie in $[\frac{\epsilon}{K}(n+1)^{-(d-3)}, (n+1)\epsilon]$, so $|\delta_n| \le H(n+1)$. Let $a_n = \mathrm{LM}(v_n, q_n)$, $b_m = \mathrm{LM}(T_m, Q_m)$ for $m \ge 0$, and $D_m = \sum_{i=m}^{m+d-2} \delta_i$. Then $v_n - q_n = a_n\delta_n$ and $T_m - Q_m = b_m D_m$, and subtracting the definition of r_n from (8.1) gives

$$a_n\delta_n + \sum_{m=\max(0,n-d+2)}^{n} b_m D_m = -r_n, \qquad n \ge 0. \tag{8.4}$$

By Lemmas 8.1 and 8.2, $0 < b_m \le (m+1)\epsilon$, and by (8.2),

$$a_n \ge \frac{q_n}{1+|\delta_n|} \ge \frac{q(1)(n+1)^{1/(d-1)}}{1+H(n+1)}. \tag{8.5}$$

Let h be real with finite support. Multiplying (8.4) by $h_n^2\delta_n$, summing over n, collecting the terms with the same b_m, and using (8.3), we obtain

$$\sum_n a_n h_n^2\delta_n^2 + \sum_m b_m\Big(\sum_{i=m}^{m+d-2} h_i\delta_i\Big)^2 = -\sum_n r_n h_n^2\delta_n - \sum_m b_m \sum_{m\le i<j\le m+d-2} (h_i - h_j)^2\delta_i\delta_j. \tag{8.6}$$

Let $p = 1/(4(d-1))$, $w_n = (n+1)^{-p}$, and, for an integer $R \ge 1$, $h_n = (w_n - w_R)_+$, which is supported on $n < R$. Then $h \le w$ and, for $m \le i < j$, $|h_i - h_j| \le |w_i - w_j| \le p(j-i)(m+1)^{-p-1}$. By Lemma 8.2, the first term on the right of (8.6) is at most $B_1 = 4\epsilon(d-1)^4\sum_{t\ge 1} t^{-1-2p}H(t)$. Using $2|\delta_i\delta_j| \le \delta_i^2 + \delta_j^2$, $|\delta_i| \le H(m+d-1)$ for $i \le m+d-2$, and the fact that a window has fewer than $(d-1)^2/2$ pairs, each with $(j-i)^2 \le (d-1)^2$, the second term is at most $B_2 = \frac{1}{2}\epsilon p^2(d-1)^4\sum_{m\ge 0}(m+1)^{-1-2p}H(m+d-1)^2$. Both series converge, and $B = B_1 + B_2$ depends only on d and ϵ. Dropping the squares on the left of (8.6), $\sum_n a_n h_n^2\delta_n^2 \le B$ for every R. As $R \to \infty$, $h_n \uparrow w_n$, and monotone convergence gives $\sum_n a_n w_n^2\delta_n^2 \le B$. By (8.5) and $2p = 1/(2(d-1))$,

$$a_n w_n^2 \ge \frac{q(1)(n+1)^{1/(2(d-1))}}{1+H(n+1)} \ge \frac{q(1)}{G}, \qquad G = \sup_{t\ge 1}\frac{1+H(t)}{t^{1/(2(d-1))}} < \infty,$$

so $\sum_n \delta_n^2 \le BG/q(1) =: M^2$. Finally, $(d-1)q(t)^{d-1} = \epsilon t - q(t)$ and $q(t)/t \to 0$ give $q(t)\big((d-1)/(\epsilon t)\big)^{1/(d-1)} \to 1$. ∎

Proof of Theorem 1.3. The asymptotic statement is part of Proposition 8.3. Let v and v' be positive solutions, with products T_m and T'_m, and put $\eta_n = \log(v'_n/v_n)$. By Proposition 8.3, $\eta \in \ell^2$. With $A_n = \mathrm{LM}(v_n, v'_n) > 0$, $B_m = \mathrm{LM}(T_m, T'_m) \le (m+1)\epsilon$ and $E_m = \sum_{i=m}^{m+d-2} \eta_i$, subtracting the two copies of (8.1) gives $A_n\eta_n + \sum_{m=\max(0,n-d+2)}^{n} B_m E_m = 0$, and the argument leading to (8.6), now without a source term, gives

$$\sum_n A_n h_n^2 \eta_n^2 \le \sum_m B_m \sum_{m \le i < j \le m+d-2} (h_i - h_j)^2 |\eta_i \eta_j|$$

for finitely supported h. Take $h_n = 1$ for $n \le R$, $h_n = 2 - n/R$ for $R < n < 2R$, and $h_n = 0$ for $n \ge 2R$. Then $|h_i - h_j| \le |i-j|/R$, and only windows with $m < 2R$ contribute, for which $B_m \le 2R\epsilon$. A site lies in at most $d-1$ windows, and within a window $\sum_{j \ne i}(j-i)^2 \le (d-1)^3$. Hence

$$\sum_n A_n h_n^2 \eta_n^2 \le \frac{2R\epsilon}{R^2} \cdot \frac{(d-1)^4}{2} \|\eta\|_{\ell^2}^2 = \frac{\epsilon(d-1)^4}{R} \|\eta\|_{\ell^2}^2.$$

For fixed n and $R \ge n$, $h_n = 1$, so $A_n \eta_n^2 \le \epsilon(d-1)^4 \|\eta\|_{\ell^2}^2 / R$ for all large R. Hence $\eta_n = 0$ for every n, and $v = v'$. The last assertions of the theorem follow from Theorems 1.1(ii) and 1.2, since $V_n = a^{2/(d-2)} v_n$ and $\epsilon = a^{-d/(d-2)}$ carry (1.1) into (1.6). ∎

In the variables $u_n = \log v_n$, (8.1) is the coordinatewise stationarity condition of the formal energy $\sum_n \left[\mathrm{e}^{u_n} + \mathrm{e}^{u_n + u_{n+1} + \cdots + u_{n+d-2}} - (n+1)\epsilon\, u_n \right]$, whose increments along finitely supported variations are well defined and strictly convex, and (8.4) is a difference of its gradients. Convexity alone does not compare two infinite sequences. In (8.5) the convexity constant is controlled by the known approximate solution q, so it grows like $n^{1/(d-1)}$ up to a logarithm, whereas a direct comparison of two unknown solutions only yields the much weaker lower bound of Lemma 8.1.

AI-use disclosure

The proof of the case $d = 4$ of Theorem 1.1, in an earlier quartic-specific form, was developed with assistance from OpenAI Codex using GPT-5.6 Sol (`gpt-5.6-sol`) at high and extra-high reasoning effort. The extension to every d (the Gamma representation and the total-positivity argument of Sections 4 and 5) was developed with assistance from Anthropic's Claude Code using Claude Opus 5.5. The manuscript was revised, and the Lean formalization of the case $d = 4$ was produced, with assistance from Claude Code using Claude Opus 5.5, Claude Fable 5 and Claude Haiku 5.5. The proof of Theorem 1.3, by comparison with an approximate solution, and the weighted energy estimate of an earlier proof for $d = 4$ were obtained in consultations with OpenAI's GPT-6 Pro, which was also used for independent referee-style checks of the proofs, the citations and the Lean statements. These systems were used as interactive research assistants to develop and stress-test the arguments, audit citations and calculations, write the formal proofs, and revise the manuscript. The author directed the research, selected and checked the final arguments, and takes responsibility for all mathematical claims, citations, computations, and wording.

Code availability

The Lean 4 formalization of the case $d = 4$ of Theorems 1.1 and 1.3 and of Theorem 1.2 is in the directory `lean` of release `v2.1` of the repository github.com/skypher/painleve (Lean 4.33.0, Mathlib commit `db584cd6d46c`). It follows the quartic arguments of the version of this paper in release `v1.1`. The statements `Painleve.Stmt_main` (the case $d = 4$ of Theorem 1.1 with Theorem 1.2) and `Painleve.Stmt_unique` (the case $d = 4$ of Theorem 1.3) are proved by `Painleve.main_theorem` and `Painleve.unique_theorem`. These proofs use only the axioms `propext`, `Classical.choice` and `Quot.sound`; the repository README gives the build commands and the file `lean/Check.lean` that checks both. The directory `runs/general-d` contains numerical checks of Sections 3–6 for $d \leq 8$, and `runs/uniq` the numerical scans of initial values used as a sanity check for Theorem 1.3.

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